Machines for Babbage's Hall

Every machine considered, in the order the hall walks them · 14 November 2026

Every non-electronic computing machine worth considering for the hall, from the list we began with and its additions (the conversation as exported, in *Mechanical computation machines*), the three papers (*Early Computing Machines*, *Computing after Babbage*, *Ball-and-disk integrator*) and the rest of the field. Clocks are left out (they are a collection of their own), and so are machines that compute with relays, valves or flowing water (see the end).

Each is scored from 1 to 5 for **interest** (how much there is to see and understand: surprise, motion, history) and for **complexity** (how hard to simulate in bpp and to print and build).

The order (since 2026-11-14) is by what a machine asks of a visitor, so that each bay builds on the ones before it, and it is the order the hall's bays are walked in, from the entrance (1) to the far end. It weighs three things: the idea the machine computes (one bit, then arithmetic with a carry, chance, number theory, programs, calculus, whole systems), how much mechanism it takes to do it, and how much the machine shows (how many things it does, how much you can try). A machine richer in its idea, or in its mechanism for the same idea, comes later. The machines fall in five tiers:

1. **One idea, little mechanism:** the abacus to the petrol-pump computer, the Domino Half Adder among them (it does one thing once: a one-bit sum).
2. **Arithmetic with a carry, or one analogue element:** the Differential, the calculators from Schickard's to Odhner's, the planimeter and its kin, the simple logic toys.
3. **Chance, number theory, programs:** the Galton and splitter boards, Jacquard's reader, the Totalisator, Lehmer's sieve, the Digi-Comp II.
4. **Analysis, dense calculators, ciphers:** Kelvin's machines, the analysers and solvers, the Millionaire, the Curta, the Hagelin, the Enigma.
5. **Whole systems:** the Differential Analyser, the fire-control computers, the Antikythera mechanism, the Z1, the difference engines, Babbage's at the very end.

Within a tier the order is judgement, and a few calls are open: the integraph sits by the planimeter, its near kin; the Digi-Comp I, the marble adders and Dr. Nim close tier 2, leading to the Digi-Comp II; the Antikythera mechanism could stand earlier, but it is the most striking machine on the list. The four at the very start (the abacus, the slide rule, Napier's bones, the lock) have little or nothing to simulate.

Until 2026-11-14 the list was ranked by interest alone, the most interesting first, and the hall walked it backwards. That put the Domino Half Adder sixth, with the Pascaline and the Galton board (all scored 4), and Lehmer's sieve second, for its modest mechanism, though factoring by modular wheels asks far more of a visitor than adding does.

Changes (2026-11-14): the list is in walking order (below); the splitter board is added (built); the Domino Half Adder is scored 3 for interest, not 4 (it does one thing once).

Changes since the first list: the Galton board, built, is scored 2 for complexity, not 1: feeding the balls in without a jam, and without a lean, took a pair of metering wheels (see its FINDINGS).

Earlier changes: Lehmer's sieve was scored 5 there (with a ball that drops when the holes line up), then 3; it is 4 here. The Curta and the Enigma were left out there (too dense; electromechanical) and are in here: the Curta as a cutaway, the Enigma for its mechanical stepping.

Status: *built*: in the hall, with its STL files (*built (accumulator)*: its first part); *bay*: has a covered bay "in preparation"; blank: not yet planned.

The list, in walking order

#	Tier	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
1	1	Abacus, suanpan	ancient	beads	nothing to simulate	1	1	
2	1	Slide rule	1620s	multiplies by adding logarithms	one slide; nothing to simulate	2	1	
3	1	Napier's bones, Genaille–Lucas rulers	1617 / 1891	rods laid side by side give a product to read off	no moving mechanism: printable, nothing to simulate	2	1	
4	1	Mechanical combination lock		the wheels' notches must line up: a lock that tests a number	wheels and a fence; simple	2	2	
5	1	Bead sort and spaghetti sort	modern	physical sorting: beads fall on rods, or rods are stood on end	beads and rods under gravity: easy	3	1	
6	1	Domino Half Adder	modern	dominoes falling add two bits	built: 268 dominoes on printed tiles	3	2	built
7	1	Slide adder (troncet, Addiator)	1889 / 1920	slides moved with a stylus add and carry by a hook	flat slides; the carry hook is the trick	3	2	
8	1	Linkage function generators (the Peaucellier–Lipkin inverter and others)	1864 / 1900s	bar linkages that compute a product, a square, a log; Peaucellier's computes 1/x	a few hinged bars each; a set of them would make one bay	3	2	
9	1	Three-dimensional cams	1930s	a cam shaped by two inputs gives a function of both (fire control's tables)	one shaped cam and a follower; prints well	3	2	
10	1	Petrol-pump price computer	1930s	multiplies the litres pumped by the price per litre, continuously, with variable-ratio gearing	a variator and counters; small	3	2	
11	2	The Differential	1827	gears that add and subtract (Pecqueur's patent; older in clocks)	built: printed bevel gears	3	2	built
12	2	Schickard's calculating clock	1623	the first known calculator: adding wheels and Napier's bones	a one-tooth carry gear (it jams on long carries: worth showing beside the Pascaline)	3	3	
13	2	Pascaline	1642	adds, with Pascal's falling-weight carry	built: three digits, sautoirs, detents, stylus and stop	4	3	built
14	2	Comptometer	1887	key-driven adder: press a key and the column moves at once	keys, sectors and carry; no crank	4	3	
15	2	Leibniz's stepped reckoner	1694	the first four-function calculator: the stepped drum	the stepped drum is the heart of the next three machines	4	3	

#	Tier	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
16	2	Thomas Arithmometer	1820	the first mass-produced calculator: stepped drums, a sliding carriage	a stepped drum per digit and a carry; a fine follow-on to the Pascaline	4	3	
17	2	Odhner pinwheel calculator (Brunsviga, Facit)	1873	pinwheels with retractable pins set the figures; turn the crank	flat pinwheels print well; the carry is the work	4	3	
18	2	Planimeter (Amsler's polar)	1856	trace a shape's outline and a rolling wheel reads its area	one wheel rolling and slipping on paper; small and clear	4	2	bay
19	2	Integraph	1880s	follows a curve and draws its integral	a wheel steering by its knife edge; a small step on from the planimeter	4	3	
20	2	Ball-and-disk integrator	1876	James Thomson's: a ball between a turning disc and a cylinder; the cylinder turns as the integral	the ball must roll without slipping; the integrator the tide predictor's brother and the analyser's ancestor	4	2	
21	2	"Basic Mechanisms" kit	1953	the US Navy fire-control course's set: differential, cam function generator, slide multiplier, component solver	each a small exhibit; together, one bay that explains the aiming machines	4	2	
22	2	Digi-Comp I	1963	plastic sliders and pegs: three flip-flops and programmable logic	slides, pegs and a lever: simple parts, many of them	4	2	
23	2	Marble adders (Turing Tumble, Wandel's)	modern	marbles flip levers to add binary numbers	contact and gravity; very like the Digi-Comp II	4	2	
24	2	Dr. Nim	1965	a marble machine that plays Nim, and wins	a few flip-flops and marbles: a little cousin of the Digi-Comp II	4	2	
25	3	Galton board	1873	balls falling through pins pile up into the normal curve	built: 150 steel balls, twelve rows of printed pins, thirteen bins; the balls metered in one at a time by two mirror-image slotted wheels (a neck jams in two dimensions, or floods the pins; one wheel made the board lean its way); first scored 1 for complexity, it took more than balls and pins	4	2	built
26	3	Splitter board	2026	a Galton board whose pins are made independent coins (a neck over every pin), with walls that swallow balls: first passage, the reflection principle, gambler's ruin, exactly	built: the Galton board's machine, twelve rows of splitters (a steel pin carrying a printed tooth), exit blocks, ramps and a tube for each row a ball can be caught at; the teeth come in sets matched to the balls' bounce (SPLITTERS.md)	4	3	built
27	3	Jacquard's card reader	1804	punched cards choose which hooks lift: the first stored program	needles, cards and hooks; a reader for a programmable exhibit	3	3	

#	Tier	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
28	3	Automatic Totalisator	1913	racetrack betting: a great shaft adder of differentials counts tickets and sets the odds	differentials (built) chained; the first real-time computer, the Science Museum says	4	3	
29	3	Lehmer's number sieve	1926 / 1932	factors numbers: loops of different lengths (bicycle chains, later toothed wheels) turn until gaps line up, and a ball or pin drops through to stop it	built: four Geneva-driven wheels of 9, 8, 7 and 5 places with pegged holes, and a cam-lifted probe; factors by Fermat's way	4	3	built
30	3	Digi-Comp II	1965	marbles rolling down a board of flip-flops count, add and multiply in binary	built: a two-level board, thirteen printed rockers tipped over by the balls alone (a steel ball in each for weight, a foot below for CLEAR) make the accumulator, MQ and distributor; hand-set switches (MULTIPLY, CLEAR, COUNT, the memory), COMPLEMENT's shutters, OVERFLOW's; a feeder held by an escapement; AUTO, a thread from a trip lever at the bottom to the START lever; programs run line by line	5	3	built (count, add, multiply, subtract, divide; AUTO)
31	4	Kelvin's tide predictor	1872	adds ten or more sine waves (the Sun's and Moon's tides) and draws a year of tides	cranks and slotted yokes are easy; the cord over the pulleys that adds them: bpp has no rope, but a rope of linked beads works if it isn't much lighter than its weights (tested: see below)	5	3	bay
32	4	Kelvin's harmonic analyser	1878	integrators find the Fourier coefficients of a measured curve (the tide predictor in reverse)	ball-and-disk integrators in a row	4	3	
33	4	Michelson and Stratton's harmonic analyser	1898	80 springs sum and analyse waves; it showed the Gibbs overshoot	springs and levers: Bullet springs, many of them	4	3	
34	4	Kelvin's equation solver (and Wilbur's)	1878 / 1936	tilting plates on pulleys and cords solve simultaneous linear equations	plates, pulleys, cords: the cords are the hard part	4	3	
35	4	Torres Quevedo's algebraic machines	1890s	find the roots of polynomials with his "endless spindle" logarithmic gearing	special gearing; little known and striking	4	4	
36	4	Millionaire	1893	the first direct multiplier: a "multiplication body" of tongues gives each product at once	complex, dense	3	5	
37	4	Curta	1948	a pocket four-function calculator: one stepped drum, nines' complements for subtraction	small and dense: the stepped drum and carry are doable, the whole machine very hard; best as an oversized cutaway (first left out as too dense)	5	4	

#	Tier	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
38	4	Hagelin M-209	1940	a fully mechanical cipher machine: six pin wheels and a drum of 27 lugged bars	all mechanical, but many small parts: pins, lugs, a print wheel	4	5	
39	4	Enigma (its rotors' stepping)	1923	a keypress steps the rotors by pawls and notches; the current's path through them lights the cipher letter	the stepping is mechanical (the middle rotor's double step comes of itself); the wiring is a lookup in Lua; prints, with wire, contacts and LEDs (first left out as electromechanical)	5	3	
40	5	Differential Analyser	1931	wheel-and-disc integrators, wired in a loop, solve differential equations and draw the answer	rolling contact that must not slip; Bush's torque amplifiers can be motors; Hartree built one in Meccano	5	4	bay
41	5	Argo clock and Dreyer table	1912 / 1911	WWI battleship fire control: plotting the target's course and range by integrators and a plotting table	integrators (as the ball-and-disk) and a plotting pen	4	4	
42	5	Torpedo Data Computer	1940s	a submarine's analogue fire control	like the rangekeeper, smaller; partly electrical	4	5	
43	5	Ford Rangekeeper Mk I	1917	a battleship's fire-control computer: integrators, differentials and cams solve where the target will be	its parts each are doable (the Differential, integrators, cams); together, a big job	5	5	
44	5	Norden bombsight	1930s	a gyro-steadied sight whose analogue computer (rate discs, cams) solves the bombing problem	the computing (disc integrators and cams) is doable; the gyro too, as in Euler's Gallery; whole, very complex	5	5	
45	5	Antikythera mechanism	c. 100 BCE	a gear train for the Sun, Moon and planets, the lunar phase, eclipse dials; a pin-and-slot gear for the Moon's varying speed	gears, as in the Differential (convex-hull teeth), and many of them; printed replicas exist	5	4	
46	5	Zuse Z1	1938	the first freely programmable computer, binary floating point, all mechanical: sliding steel plates and pins	its plate logic is lovely to simulate gate by gate; the whole machine is far beyond us	5	5	
47	5	Scheutz difference engine	1853	the first working difference engine, with a printer	as Babbage's; one of them is enough	3	5	
48	5	Babbage's Difference Engine	1822 / 1849	tabulates a polynomial by adding its differences, with no multiplying	two orders of difference, three digits, is a good model: figure wheels and Babbage's carry, after the Pascaline	5	5	bay

By kind

- **Adding and counting** (digital): Schickard, the Pascaline, Leibniz, the Arithmometer, the pinwheels, the Comptometer, the Millionaire, the Curta, the slide adder, the Totalisator, the petrol-pump computer.
- **Tabulating and programmed:** the Difference Engines (Babbage's, Scheutz's), Jacquard's reader, the Zuse Z1.
- **Logic and play:** the Digi-Comps, Dr. Nim, marble adders, the domino adder, the Galton board, the bead and spaghetti sorts.
- **Number theory, ciphers and locks:** Lehmer's sieve, the Enigma, the Hagelin M-209, the combination lock.
- **Calculus and waves** (analogue): the planimeter, the integraph, the ball-and-disk integrator, Kelvin's tide predictor and harmonic analyser, Michelson and Stratton's analyser, the Differential Analyser.
- **Algebra:** Kelvin's and Wilbur's equation solvers, Torres Quevedo's root finders, linkage function generators, three-dimensional cams, the "Basic Mechanisms" kit.
- **Aiming:** the Argo clock and Dreyer table, the rangekeeper, the Norden bombsight, the Torpedo Data Computer.
- **Astronomy:** the Antikythera mechanism.

Left out, and why

- **Electromechanical and electronic:** Stibitz's Model K (relays), the bombes, the Harvard Mark I, Colossus, ENIAC and after, and the Mallock machine (electrical). The Enigma is kept because its stepping is mechanical and its wiring is a fixed lookup; the bombe is not, being relays and motors.
- **Water:** Lukyanov's water integrator and Phillips's MONIAC compute with flowing water, which Bullet does not do (rigid and soft bodies only).
- **Clocks and orreries:** a collection of their own.
- **Program stores that don't compute:** music boxes and Jacquard's looms (Jacquard's card reader is kept, as an input for another machine).
- **Hollerith's machines** (electrical counting) and **MEMS and nanomachines** (too small to print).

How the hall uses this list

The hall's bays are walked in the list's order, so a new exhibit goes in its place in it and the bays after it move along one (the hall grows by a pair of facing bays when it needs one). With eleven bays (since 2026-11-14) the hall holds the seven built and the four covered "in preparation", in this order: the Domino Half Adder and the Differential (west 1, east 1), the Pascaline and the planimeter (west 2, east 2), the Galton board and the splitter board facing each other (west 3, east 3), Lehmer's sieve and the Digi-Comp II (west 4, east 4), the tide predictor and the Differential Analyser (west 5, east 5), the Difference Engine (west 6). The machines are to be built from the start of the list on.

Rope. bpp has no rope (its soft bodies are cloth). A rope of small spheres joined by ball joints works, tested as an Atwood machine over a 5 cm pulley: with 60 beads making a 50 g rope between 200 g and 250 g weights, it stretched under 0.3% and the weights accelerated at 95 to 103 cm/s^2 against the textbook 98, at 2400 steps a second and 100 solver passes, about a second of computing per simulated second. A rope 2500 times lighter per bead than its weights broke at any step size: the rope must not be too much lighter than what hangs on it. That suits the tide predictor (its pen weight can be light) and the equation solvers. The other way is to add Bullet's own rope (`btSoftBodyHelpers::CreateRope`) to bpp, a

C++ change. The best candidates for new bays, for interest against effort, were the Galton board and the planimeter (quick and clear), and the Enigma and the Antikythera mechanism (striking, and within reach). The Galton board is built (2026-11-12) and the planimeter has its bay.